

ORIGINAL RESEARCH

HEART FAILURE

Seated Pulmonary Artery Pressure Management in Patients With Heart Failure



12-Month Outcomes of the PROACTIVE-HF Trial

Jason L. Guichard, MD,^a Eric L. Bonno, MD,^a Michael E. Nassif, MD,^b Taiyeb M. Khumri, MD,^b David Miranda, MD,^c Orvar Jonsson, MD,^d Hirak Shah, MD,^e Tamas Alexy, MD,^f Gregory P. Macaluso, MD,^g James Sur, MD,^g Gavin Hickey, MD,^h Patrick McCann, MD,ⁱ Jennifer A. Cowger, MD,^j Amit Badiye, MD,^k Wayne D. Old, MD,^k Yasmin Raza, MD,^l Luke Masha, MD,^m Chandra Kunavarapu, MD,ⁿ Mosi Bennett, MD,^c Faisal Sharif, MD,^o Michael Kiernan, MD,^p Wilfried Mullens, MD,^{q,r} Sandra V. Chaparro, MD,^s Claudius Mahr, MD,^t Rohit R. Amin, MD,^u Nicholas J. Hiivala, BME,^v Max M. Owens, PhD,^v Andrea Sauerland, MS,^v Omid Forouzan, PhD,^v Liviu Klein, MD, MS^w

ABSTRACT

BACKGROUND In the PROACTIVE-HF (A Prospective, Multi-Center, Open Label, Single Arm Clinical Trial Evaluating the Safety and Efficacy of the Cordella Pulmonary Artery Sensor System in NYHA Class III Heart Failure Patients) trial, remote heart failure (HF) management using seated mean pulmonary artery pressure (mPAP) and vital signs was safe and resulted in a low rate of HF hospitalizations and mortality through 6 months.

OBJECTIVES The authors evaluated the effect of managing seated mPAP with the Cordella system on outcomes in patients with HF through 12 months.

METHODS In a single-arm, open-label trial, conducted in 75 European and U.S. centers, the authors enrolled HF patients with NYHA functional class III symptoms, irrespective of ejection fraction, and recent HF hospitalization and/or elevated natriuretic peptides. The prespecified, powered, secondary effectiveness endpoint at 12 months required the HF hospitalization or all-cause mortality rate to be lower than a performance goal of 0.70 events/patient/12 months, established from previous hemodynamic monitoring trials. Device/system-related complications, pressure sensor failure, and serious adverse events were examined.

RESULTS Between February 7, 2020, and March 31, 2023, 456 patients were implanted in a modified intent-to-treat cohort. The 12-month event rate was 0.36 (95% CI: 0.31-0.42), which was significantly lower than the performance goal (0.36 vs 0.70; $P < 0.0001$). There were no device/system-related complications or pressure sensor failures beyond the 6-month primary results (0.8% and 0.2%, respectively).

CONCLUSIONS Remote HF management using seated mPAP and vital signs in NYHA functional class III patients is safe and results in a low rate of HF hospitalizations and mortality over 12 months. These results support the use of seated mPAP monitoring and extend the evidence that pulmonary artery pressure-guided management improves HF outcomes. (A Prospective, Multi-Center, Open Label, Single Arm Clinical Trial Evaluating the Safety and Efficacy of the Cordella Pulmonary Artery Sensor System in NYHA Class III Heart Failure Patients [PROACTIVE-HF Trial]; [NCT04089059](https://clinicaltrials.gov/ct2/show/study/NCT04089059)) (JACC Heart Fail. 2026;14:102847) © 2026 by the American College of Cardiology Foundation.

**ABBREVIATIONS
AND ACRONYMS****6MWT** = 6-minute walk test**GDMT** = guideline-directed medical therapy**HF** = heart failure**HFmrEF** = heart failure with mid-range ejection fraction**HFpEF** = heart failure with preserved ejection fraction**HFrEF** = heart failure with reduced ejection fraction**IV** = intravenous**KCCQ** = Kansas City Cardiomyopathy Questionnaire**LVEF** = left ventricular ejection fraction**mPAP** = mean pulmonary artery pressure**NT-proBNP** = N-terminal pro-B-type natriuretic peptide**PA** = pulmonary artery**PAP** = pulmonary artery pressure**SGLT2** = sodium-glucose cotransporter 2

Increasing cardiac filling pressures are a hallmark of symptomatic chronic heart failure (HF) across all ejection fractions and are strongly associated with HF events and cardiovascular mortality.^{1,2} Despite significant advances in neurohormonal and device therapies for heart failure with reduced ejection fraction (HFrEF), hospitalized patients have a 50% risk of rehospitalization within 6 months of discharge,³ along with persistent symptoms that undermine quality of life and sense of control over their disease. Although these challenges are equally prevalent among patients with heart failure with preserved ejection fraction (HFpEF), who represent half of HF hospitalizations, the therapies that have been effective in decreasing HF hospitalizations in HFrEF have shown limited benefit in patients with HFpEF.^{1,4}

Remote monitoring of pulmonary artery pressures (PAPs) with active medical intervention to maintain PAPs within a target range has been shown to be an effective strategy for reducing HF hospitalizations in both HFrEF and HFpEF, and reducing mortality in HFrEF, in patients with NYHA functional class III symptoms with a prior HF hospitalization.^{5,6}

The Cordella PA (Pulmonary Artery) Sensor System combined with the Cordella HF System (Cordella, Endotronix Inc) provides comprehensive

clinical assessment of patients with HF at home including either seated or supine PAP (systolic, diastolic, and mean), body weight, blood pressure, heart rate, oxygen saturation, and self-reported symptoms, and has been shown to be safe and to markedly reduce HF hospitalizations in the 6 months after implantation.⁷

The PROACTIVE-HF (A Prospective, Multi-Center, Open Label, Single Arm Clinical Trial Evaluating the Safety and Efficacy of the Cordella Pulmonary Artery Sensor System in NYHA Class III Heart Failure Patients) trial was originally approved in 2019 as a prospective, randomized, controlled, single-blind, multicenter clinical trial to evaluate the safety and effectiveness of Cordella in patients with HF with NYHA functional class III symptoms with either prior HF hospitalization and/or elevated natriuretic peptides.^{7,8} In December 2021, after 160 patients had been randomized, following consultation with the Food and Drug Administration, PROACTIVE-HF was changed to a single-arm, open-label trial with blinded endpoint assessment and prespecified safety and effectiveness endpoints defined from previous hemodynamic monitoring trials, to provide objective evidence of a similar risk-benefit profile to a commercially available device in patients with NYHA functional class III. PROACTIVE-HF met all primary safety, primary efficacy, and secondary endpoints at 6 months.⁷

This report extends these analyses over a 12-month period and examines the outcomes across the ejection fraction spectrum.

From the ^aDepartment of Medicine, Division of Cardiology, Section for Advanced Heart Failure, Pulmonary Hypertension, and Mechanical Circulatory Support, Prisma Health-Upstate, Greenville, South Carolina, USA; ^bSaint Luke's Mid-American Heart Institute, Kansas City, Missouri, USA; ^cDepartment of Cardiology, Minneapolis Heart Institute, Minneapolis, Minnesota, USA; ^dDepartment of Cardiology, North Central Heart and Avera Heart Hospital, Sioux Falls, South Dakota, USA; ^eDepartment of Cardiology, University of Kansas Medical Center, Kansas City, Kansas, USA; ^fDepartment of Cardiology, University of Minnesota Medical Center, Minneapolis, Minnesota, USA; ^gDepartment of Cardiology, Advocate Christ Medical Center, Oak Lawn, Illinois, USA; ^hDepartment of Cardiology, University of Pittsburgh Medical Center, Pittsburgh, Pennsylvania, USA; ⁱDepartment of Cardiology, Prisma Health, Columbia, South Carolina, USA; ^jDivision of Cardiovascular Medicine, Henry Ford Hospital, Detroit, Michigan, USA; ^kDepartment of Cardiology, Sentara Healthcare, Norfolk, Virginia, USA; ^lDepartment of Cardiology, Northwestern, Chicago, Illinois, USA; ^mDepartment of Cardiology, Oregon Health and Science University, Portland, Oregon, USA; ⁿDepartment of Cardiology, Methodist Healthcare, San Antonio, Texas, USA; ^oDepartment of Cardiology, Galway University Hospital, Saolta Group, CURAM and University of Galway, Galway, Ireland; ^pCardiovascular Center, Division of Cardiology, Tufts Medical Center, Boston, Massachusetts, USA; ^qDepartment of Cardiology, Ziekenhuis Oost-Limburg, Genk, Belgium; ^rHasselt University, Hasselt, Belgium; ^sMiami Cardiac and Vascular Institute, Division of Cardiology, Baptist Health South Florida, Miami, Florida, USA; ^tInstitute for Advanced Cardiac Care, Medical City, Dallas, Texas, USA; ^uDepartment of Cardiology, Ascension Sacred Heart Hospital, Pensacola, Florida, USA; ^vClinical Science, Endotronix Inc, Naperville, Illinois, USA; and the ^wAdvanced Heart Failure Comprehensive Care Center and Division of Cardiology, University of California-San Francisco, San Francisco, California, USA.

Biykem Bozkurt, MD, PhD, served as acting Editor-in-Chief and main adjudicator for this paper.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

METHODS

STUDY DESIGN AND PARTICIPANTS. The design and rationale for the change from randomized to single-arm trial was previously published.⁸ Briefly, under the randomized design, patients in the treatment arm were instructed to take their PAP measurements daily (not visible to the patient for 12 months) in addition to their weight, blood pressure, heart rate, and oxygen saturation (visible to patient and clinician), which were wirelessly transmitted to a secure website for clinicians to review. In the treatment arm, a patients' daily PAP and vital sign trends were visible, which enabled clinicians to proactively manage patients to a target seated mean pulmonary artery pressure (mPAP) per protocol using guideline-directed medical therapy (GDMT), vasodilators, and diuretics.

Patients in the control arm were instructed to take their PAP measurements daily (not visible to the patient nor clinician for 12 months) in addition to their weight, blood pressure, heart rate, and oxygen saturation (visible to patient and clinician), which were wirelessly transmitted to a secure website for clinicians to review. In the control arm, only patients' daily vital sign trends were visible, which enabled clinicians to proactively manage patients using GDMT, vasodilators, and diuretics.

Following the change to a single-arm design, patients in the treatment arm (now former treatment arm) were informed of their original assignment but continued the same protocol (daily PAP and vital sign measurements with only clinician access to PAP). Patients in the control arm (now former control arm) were informed of their original assignment and both patients and clinicians were unblinded to PAP data, regardless of when they received the sensor implant. These patients did not contribute to the safety and effectiveness endpoints and their results post unblinding have been reported on separately.⁹

Following the change to the single-arm design, newly implanted single-arm patients followed the same protocol as the former treatment arm (daily PAP and vital sign measurements with only clinician access to PAP for 12 months). Patients in the former treatment arm, along with newly implanted single-arm patients, comprised the primary cohort and contributed to the safety and effectiveness endpoints of PROACTIVE-HF.

The study was undertaken in accordance with the Declaration of Helsinki and approved by the relevant competent authorities and independent ethics committees. All patients provided written informed

consent. Eligible patients for PROACTIVE-HF were men or women >18 years of age with a diagnosis and treatment of HF (regardless of ejection fraction) for at least 3 months and NYHA functional class III symptoms at the time of screening. Patients were to be on stable, optimally titrated GDMT for at least 30 days before screening. Patients had to have at least 1 HF hospitalization, HF treatment in a hospital daycare setting, or urgent outpatient clinic HF visit within 12 months and/or elevated N-terminal pro-B-type brain natriuretic peptide (NT-proBNP). Elevated NT-proBNP was defined as NT-proBNP $\geq 1,500$ pg/mL for patients with a left ventricular ejection fraction (LVEF) $\leq 50\%$ and NT-proBNP ≥ 800 pg/mL for patients with an LVEF $> 50\%$, with threshold correction for body mass index using a 4% reduction per body mass index unit over 25 kg/m² (see [Supplemental Methods](#), Inclusion Criteria). A full list of inclusion and exclusion criteria can be found in the [Supplemental Methods](#).

TRIAL PROCEDURES. Following implantation of the Cordella PAP sensor,¹⁰ patients in the primary cohort (former treatment arm and newly enrolled single-arm patients) were treated to a target trend seated mPAP of 5 to 20 mm Hg,¹¹ achieved through titration of GDMT, vasodilators, and diuretic agents via a trial-specific treatment guideline ([Supplemental Methods](#)). Trend seated mPAP was defined as a 7-day moving average of daily mPAP readings.

All patients had follow-up visits at 1, 6, and 12 months for clinical assessment, with a 3-month follow-up visit performed as a remote/virtual telemonitoring visit. All patients will participate in the study for 5 years with visits at 18, 24, 36, 48, and 60 months.

OUTCOMES. The prespecified, powered secondary effectiveness endpoint was the 12-month incidence of HF hospitalization or all-cause mortality compared with a performance goal in the modified intent-to-treat population, defined as all patients in the former treatment arm and single arm who received a sensor implant. An HF hospitalization was defined as an event in which the patient was admitted to the hospital with a primary diagnosis of HF, the length of stay was at least 24 hours, the patient exhibited new or worsening symptoms of HF on presentation, had objective evidence of new or worsening HF, and received initiation or intensification of treatment specifically for HF. Key secondary outcomes include number of HF hospitalizations at 12 months post implantation compared with 12 months before implantation, combined outcome of first and recurrent

HF hospitalizations, outpatient intravenous (IV) diuretic visits, and all-cause mortality, change from baseline of PAP, vital signs, NT-proBNP, quality of life and functional capacity, echocardiogram markers, and HF medication changes, as well as patient and site compliance with data transmissions. Patient compliance was defined as transmitting clinical data at least 5 of 7 days per week (did not need to be consecutive). Site compliance was defined as acknowledgment of patient data at least twice per week with a maximum of 4 days between.

Safety endpoints include frequency of pressure sensor failure, device/system-related complications, and serious adverse events throughout 12 months.

STATISTICAL ANALYSIS. For powered secondary effectiveness success, the upper confidence bound of the observed event rate was required to be <0.70 events per 12 months, or equivalently, the corresponding P value from the hypothesis test of 12-month incidence rate compared with 0.70 was <0.025 . Furthermore, the observed incidence rate was required to be <0.59 . Incidence was calculated using Poisson regression with the exposure time as an offset term, with exposure time defined as the time from implantation to death, early study discontinuation, or 12 months. A sensitivity analysis repeated powered secondary effectiveness analysis using negative binomial regression. Powered secondary endpoint analyses were repeated in the different study arms (ie, single arm, former treatment arm, former control arm) using the same alpha threshold as was used with the full sample (0.025). Comparison of event rates between different study arms was done with Poisson regression. An alpha level of 0.15 was used for study-arm comparisons to balance type I and II error concerns given the study was not prospectively powered for a study-arm analysis. We regarded study-arm comparisons as exploratory and hypothesis generating. Other secondary effectiveness endpoints were compared at baseline and 12 months using paired sample Student's t -tests or Wilcoxon signed rank tests for continuous variables (depending on variable normality) and chi-square tests for categorical variables with alpha level of 0.05. For other secondary effectiveness endpoint comparisons, only patients with data at baseline and 12 months (paired) were used. Medication changes are reported descriptively. No data were missing from the powered secondary effectiveness endpoint. In other secondary effectiveness analyses, missing data were treated with listwise deletion.

HF hospitalization, all-cause mortality, and outpatient IV diuretic visits were analyzed using the Andersen-Gill extension of the Cox proportional hazard model with robust sandwich estimate of variance. This approach was used to examine the primary cohort and ejection fraction groups: HFpEF, HFREF, and heart failure with mid-range ejection fraction (HFmrEF). All assumptions of this model were examined and found not to be violated.

Most analyses were performed in SAS version 9.4. (SAS Institute Inc) with PAP and medication change analyses, as well as figure generating, being performed in Python version 3.9.16.

RESULTS

Between December 2019 and March 2023, 738 patients were screened with 125 screening failures. Before the change to single arm, 72 patients were randomized to the former control arm. There were 48 withdrawals before implantation, leaving 493 patients in the intent-to-treat population (safety analysis) and 37 aborted implantations, leaving 456 patients in the modified intent-to-treat population (primary cohort). This population was composed of the former treatment arm ($N = 88$) and newly enrolled single arm ($N = 368$). At 12 months, 395 patients in the primary cohort completed follow-up (Supplemental Figure 1).

There were 201 (44.1%) patients with HFpEF (LVEF $\geq 50\%$), 208 (45.6%) with HFREF (LVEF $\leq 40\%$), and 45 (9.9%) with HFmrEF (LVEF $>40\%$ - $<50\%$). Comparing the 3 groups, patients with HFpEF were older and female; had higher body mass index; were more often White; had more frequent chronic kidney disease and chronic obstructive pulmonary disease; had a shorter duration of HF; less ischemic and more hypertensive disease; had more frequent atrial fibrillation; fewer defibrillator implants; and lower estimated glomerular filtration rate, NT-proBNP, Kansas City Cardiomyopathy Questionnaire (KCCQ)-Overall Summary Score, and 6-minute walk test (6MWT). Except for angiotensin-converting enzyme inhibitors, nitrates, and diuretics, patients with HFpEF were on fewer baseline HF medications (Table 1). Results for the HFmrEF cohort fell in the intermediate range between HFpEF and HFREF.

In the first 12 months, the incidence of HF hospitalizations or all-cause mortality was 0.36 (95% CI: 0.31-0.42) events per patient, which was nearly half of the performance goal set from previous trials (0.36 vs 0.70; Poisson regression $P < 0.0001$). Given that the upper confidence bound of the observed event

TABLE 1 Baseline Characteristics

	HFpEF (EF ≥50%) (n = 201)	HFrEF (EF ≤40%) (n = 208)	HFmrEF (EF 41%-49%) (n = 45)
Demographics			
Age, y	66.4 (11.1)	60.2 (12.7)	65.8 (14.5)
Female	102 (50.7)	59 (28.4)	18 (40.0)
Body mass index, kg/m ²	38.0 (9.0)	33.8 (8.0)	35.5 (9.1)
Race			
White	166 (82.6)	143 (68.8)	36 (80.0)
Black	24 (11.9)	52 (25.0)	8 (17.8)
Asian	5 (2.5)	2 (1.0)	0 (0)
Native Hawaiian or Pacific Islander	2 (1.0)	1 (0.5)	0 (0)
American Indian/Alaskan Native	1 (0.5)	1 (0.5)	0 (0)
Other	1 (0.5)	3 (1.4)	0 (0)
Race not reported	2 (1.0)	6 (2.9)	1 (2.2)
Ethnicity			
Hispanic or Latino	5 (2.5)	14 (6.7)	1 (2.2)
Not Hispanic or Latino	177 (88.1)	181 (87.0)	41 (91.1)
Ethnicity not reported	6 (3.0)	6 (2.9)	1 (2.2)
Unknown	13 (6.5)	7 (3.4)	2 (4.4)
Medical history/surgical history			
Diabetes mellitus	77 (38.3)	66 (31.7)	18 (40.0)
Chronic kidney disease	101 (50.2)	79 (38.0)	18 (40.0)
COPD	55 (27.4)	29 (13.9)	7 (15.6)
Coronary artery bypass graft	35 (17.4)	38 (18.3)	8 (17.8)
Percutaneous coronary intervention	46 (22.9)	72 (34.6)	16 (35.6)
Valve repair/replacement	8 (4.0)	10 (4.8)	3 (6.7)
HF history			
Time since first diagnosis of HF, mo	50.5 (60.5)	68.4 (78.3)	44.8 (43.6)
Median	28.73	40.17	30.7
Minimum	3.07	2.97	3.27
Maximum	356.7	454.37	199.3
Etiology of HF			
Ischemic	40 (19.9)	88 (42.3)	15 (33.3)
Nonischemic	161 (80.1)	120 (57.7)	30 (66.7)
Other cardiovascular history			
Hypertension	188 (93.5)	173 (83.2)	40 (88.9)
Coronary artery disease	110 (54.7)	131 (63.0)	28 (62.2)
Myocardial infarction	40 (19.9)	67 (32.2)	11 (24.4)
Atrial fibrillation	121 (60.2)	91 (43.8)	22 (48.9)
Transient ischemic attack	15 (7.5)	5 (2.4)	2 (4.4)
Stroke	24 (11.9)	18 (8.7)	5 (11.1)
Peripheral vascular disease	36 (17.9)	17 (8.2)	8 (17.8)
Carotid artery disease	18 (9.0)	10 (4.8)	5 (11.1)
Ventricular tachycardia	25 (12.4)	55 (26.4)	8 (17.8)
Ventricular fibrillation	6 (3.0)	12 (5.8)	0 (0)
Atrial flutter	32 (15.9)	20 (9.6)	6 (13.3)
Cardiac resynchronization therapy	36 (17.9)	32 (15.4)	12 (26.7)
Implantable cardioverter-defibrillator	20 (10.0)	129 (62.0)	11 (24.4)
Guideline-directed medical therapy^a			
Agents acting on the renin-angiotensin system	69 (34.3)	176 (84.6)	35 (77.8)
ARNI	25 (12.4)	150 (72.1)	26 (57.8)
ARB	35 (17.4)	21 (10.1)	9 (20.0)
ACEI	12 (6.0)	8 (3.8)	1 (2.2)
Beta-blocker	139 (69.2)	188 (90.4)	37 (82.2)
Diuretic	190 (94.5)	191 (91.8)	39 (86.7)
Loop diuretic	186 (92.5)	191 (91.8)	39 (86.7)
Thiazide diuretic	41 (20.4)	23 (11.1)	5 (11.1)
Aldosterone antagonist	115 (57.2)	151 (72.6)	36 (80.0)
SGLT2 inhibitor	91 (45.3)	141 (67.8)	27 (60.0)
Hydralazine	23 (11.4)	11 (5.3)	2 (4.4)
Nitrates	35 (17.4)	41 (19.7)	5 (11.1)

Continued on the next page

TABLE 1 Continued

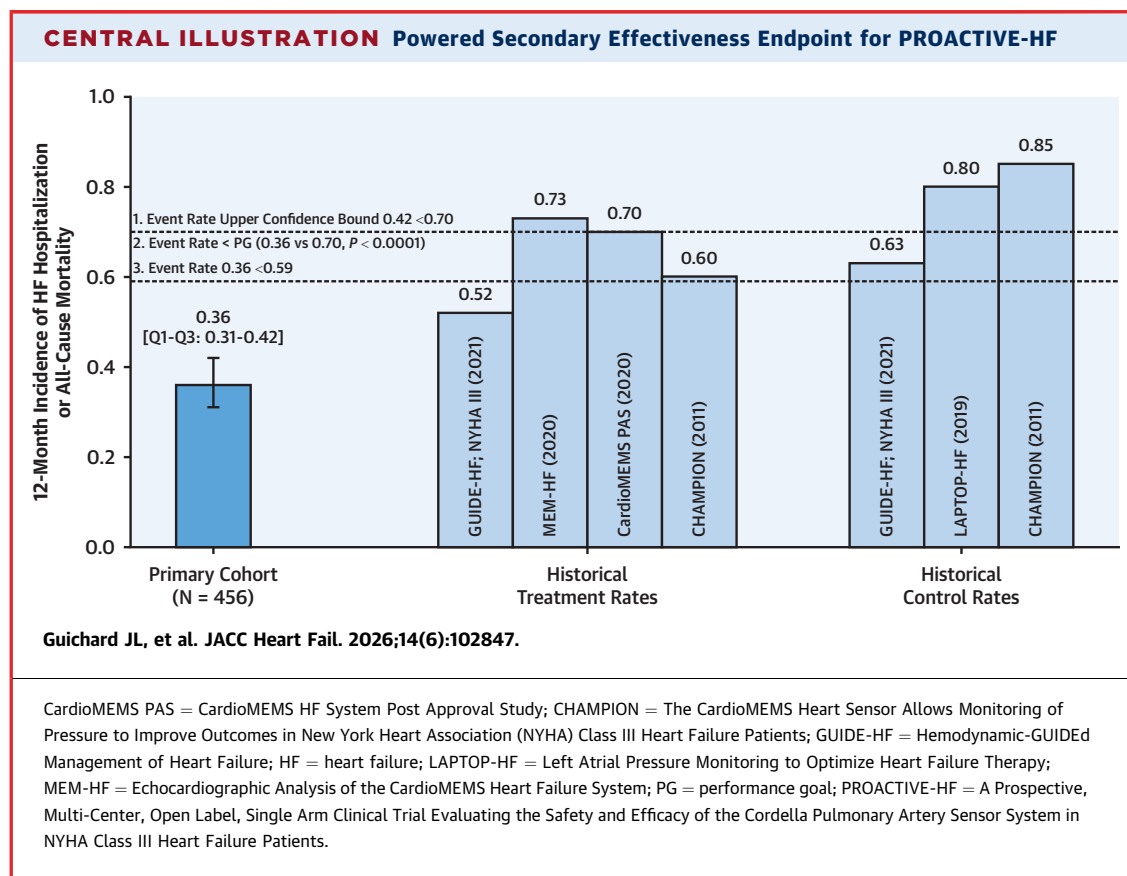
	HFpEF (EF ≥50%) (n = 201)	HFREF (EF ≤40%) (n = 208)	HFmrEF (EF 41%-49%) (n = 45)
Laboratory			
eGFR, mL/min/1.73m ²	50.9 (17.3)	58.0 (19.6)	56.5 (17.8)
Creatinine, μmol/L	128.7 (48.8)	123.4 (40.0)	116.1 (39.0)
NT-proBNP, pg/mL	1,313.7 (2,432.6)	2,166.2 (3,651.6)	1,685.6 (1,670.5)
Hemodynamics (right heart catheterization)			
Systolic pulmonary artery pressure, mm Hg	44.3 (14.4)	42.0 (15.5)	44.4 (14.4)
Diastolic pulmonary artery pressure, mm Hg	19.1 (7.3)	18.6 (8.8)	18.2 (6.1)
Mean pulmonary artery pressure, mm Hg	28.7 (9.5)	27.7 (11.0)	28.1 (8.5)
Right atrial pressure, mm Hg	10.3 (5.0)	9.0 (8.2)	10.0 (6.0)
Pulmonary capillary wedge pressure, mm Hg	16.7 (7.0)	16.7 (10.3)	14.9 (6.7)
Cardiac output, L/min	5.5 (1.5)	5.2 (4.4)	4.9 (1.4)
Cardiac index, L/min/m ²	2.5 (0.6)	2.2 (0.6)	2.3 (0.6)
Right ventricular systolic pressure, mm Hg	44.6 (14.5)	42.7 (18.2)	44.4 (13.0)
Right ventricular diastolic pressure, mm Hg	6.1 (5.5)	6.8 (11.2)	5.6 (4.8)
Right ventricular end-diastolic pressure, mm Hg	11.3 (5.2)	11.2 (13.2)	11.3 (6.2)
Vital signs			
Systolic blood pressure	128.1 (19.0)	115.1 (17.1)	123.6 (18.6)
Diastolic blood pressure	73.0 (12.8)	72.8 (12.3)	73.5 (13.8)
Heart rate	76.2 (12.9)	77.5 (14.9)	80.3 (15.4)
Blood oxygen saturation	95.3 (3.1)	96.3 (2.5)	95.8 (2.7)
Ambulatory hemodynamics during first week			
Seated systolic pulmonary artery pressure, mm Hg	33.5 (14.4)	31.7 (14.0)	33.3 (15.4)
Seated diastolic pulmonary artery pressure, mm Hg	12.4 (8.1)	12.7 (9.0)	11.8 (7.1)
Seated mean pulmonary artery pressure, mm Hg	20.6 (10.3)	19.9 (10.6)	19.9 (9.6)
Heart rate, beats/min	77.9 (14.5)	78.4 (14.9)	78.2 (13.2)
Echocardiogram			
LVEF	59.0 (6.5)	26.3 (8.2)	45.0 (2.2)
Left ventricular end-diastolic dimension	10.1 (15.4)	18.7 (36.9)	12.9 (19.9)
Functional capacity and quality of life			
NYHA functional class III	200 (99.5)	208 (100.0)	45 (100.0)
KCCQ-OSS, points	49.6 (23.1)	55.7 (21.9)	52.8 (25.0)
6MWT, m	224.7 (118.2)	289.6 (115.4)	271.2 (121.8)
Enrollment type			
HFH	55 (27.4)	79 (38.0)	14 (31.1)
NT-proBNP	42 (20.9)	34 (16.3)	13 (28.9)
HFH and NT-proBNP	104 (51.7)	94 (45.2)	18 (40.0)

Values are n (%). ^aPlease note that there were 3 subjects in the HFpEF cohort and 3 subjects in the HFREF groups that were on multiple renin-angiotensin system inhibitors.

6MWT = 6-minute walk test; ACEI = angiotensin-converting enzyme inhibitor; ARB = angiotensin II receptor blocker; ARNI = angiotensin receptor-neprilysin inhibitor; COPD = chronic obstructive pulmonary disease; EF = ejection fraction; eGFR = estimated glomerular filtration rate; HF = heart failure; HFH = heart failure hospitalization; HFmrEF = heart failure with mid-range ejection fraction; HFpEF = heart failure with preserved ejection fraction; HFREF = heart failure with reduced ejection fraction; KCCQ-OSS = Kansas City Cardiomyopathy Questionnaire-Overall Summary Score; LVEF = left ventricular ejection fraction; NT-proBNP = N-terminal pro-B-type natriuretic peptide; SGLT2 = sodium-glucose cotransporter 2.

rate was less than the performance goal (0.42 vs 0.70), the observed rate was significantly lower than the performance goal, and the observed rate was <0.59, all criteria for powered secondary effectiveness success were satisfied (**Central Illustration**). Sensitivity analysis demonstrated that results of the powered secondary effectiveness endpoint analysis were highly similar using negative binomial regression instead of Poisson regression (negative binomial event rate = 0.40 [95% CI: 0.32-0.50]; negative binomial regression *P* for comparison against

performance goal <0.0001). Events in the primary endpoint consisted of 126 first and recurrent HF hospitalizations in 87 patients (0.29 events per patient in 12 months [95% CI: 0.24-0.34 events per patient in 12 months]) and 33 all-cause deaths (0.08 events per patient in 12 months [95% CI: 0.05-0.11 events per patient in 12 months]). There was no difference between the event rates of the newly enrolled single arm (0.38; 95% CI: 0.32-0.45) and the former treatment arm (0.28; 95% CI: 0.19-0.42) (Poisson regression *P* = 0.16). The event rate in the

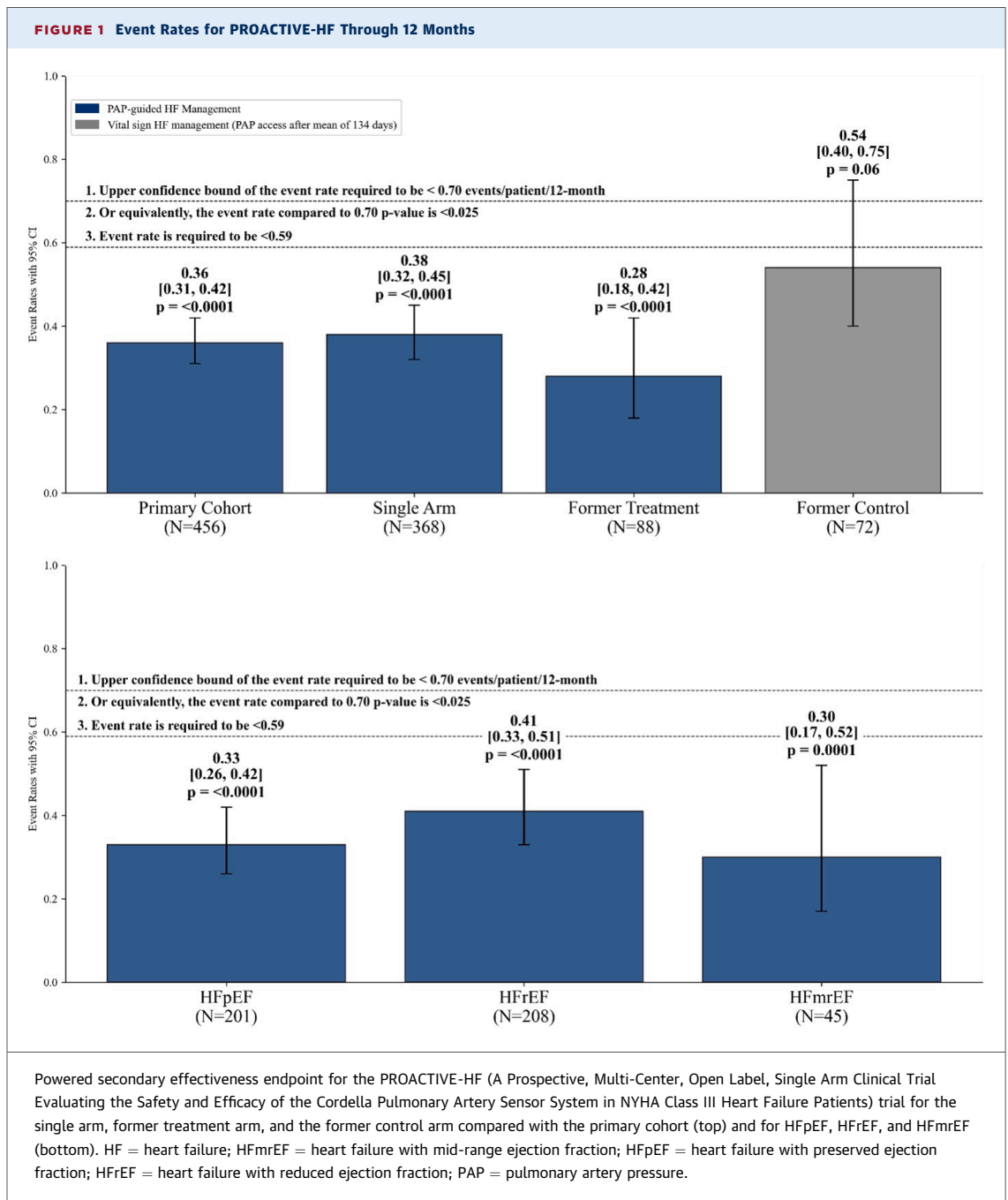


former control arm was 0.54 (95% CI: 0.40-0.75), which was not significantly reduced from the performance goal (0.54 vs 0.70; Poisson regression $P = 0.06$). Furthermore, the event rate of the former control arm was significantly higher than both the former treatment arm (0.54 vs 0.28, Poisson regression $P = 0.01$) and the primary cohort event rate (0.54 vs 0.36, Poisson regression $P = 0.03$) (Figure 1). There were 64 primary events (0.33/patient-year) in the HFpEF population, 81 primary events (0.41/patient-year) in the HFrEF population, and 13 primary events (0.30/patient-year) in the HFmrEF population. There was no difference between HFpEF and HFrEF primary event rates ($P = 0.22$).

There were 24 patients with 29 outpatient IV diuretic visits (0.06; 95% CI: 0.05-0.10) and combining outpatient IV diuretic visits with HF hospitalizations and mortality resulted in a 12-month incidence of 0.43 (95% CI: 0.37-0.50) and 12-month cumulative hazard rate of 0.40. Results were similar for HFpEF (cumulative hazard rate: 0.40) and HFrEF patients (cumulative hazard rate: 0.42) (Supplemental Figure 2). The number of HF hospitalizations, deaths, and outpatient IV diuretic events

and recurrent events for all groups are shown in Supplemental Table 1.

Comparison of key secondary endpoints from baseline through 12 months indicated significant improvement in KCCQ score of 5.7 points (52.8 ± 22.8 points vs 58.5 ± 24.4 points, paired sample Student's t -test $P < 0.0001$, paired $n = 335$, missing = 15.2%), 6MWT of 34.6 m (259.5 ± 121.1 m vs 294.1 ± 132.9 m, paired sample Student's t -test $P = 0.0004$, paired $n = 298$, missing = 24.6%), and NYHA functional class ($N = 165$ improved by at least 1 class, chi-square $P < 0.0001$, paired $n = 384$, missing = 2.8%), and NT-proBNP ($1,731.4 \pm 3,012.4$ pg/mL vs $1,463.0 \pm 3,606.0$ pg/mL, Wilcoxon signed rank $P = 0.006$, paired $n = 387$, missing = 2.0%) (Figure 2). Both HFpEF and HFrEF improved in KCCQ score (HFpEF: 50.0 ± 22.9 vs 54.9 ± 23.8 ; $P = 0.02$; HFrEF: 55.9 ± 21.6 vs 61.4 ± 24.6 ; $P < 0.001$) and NYHA functional class (HFpEF 19.9% improved; $P < 0.001$; HFrEF 24.3% improved; $P < 0.001$). For 6MWT, only HFpEF improved (224.7 ± 118.2 vs 281.2 ± 143.7 ; $P < 0.001$) and for NT-proBNP, only HFrEF improved ($2,166.2 \pm 3,651.6$ vs $1,589.0 \pm 2,671.7$; $P < 0.001$) (Supplemental Figure 3).

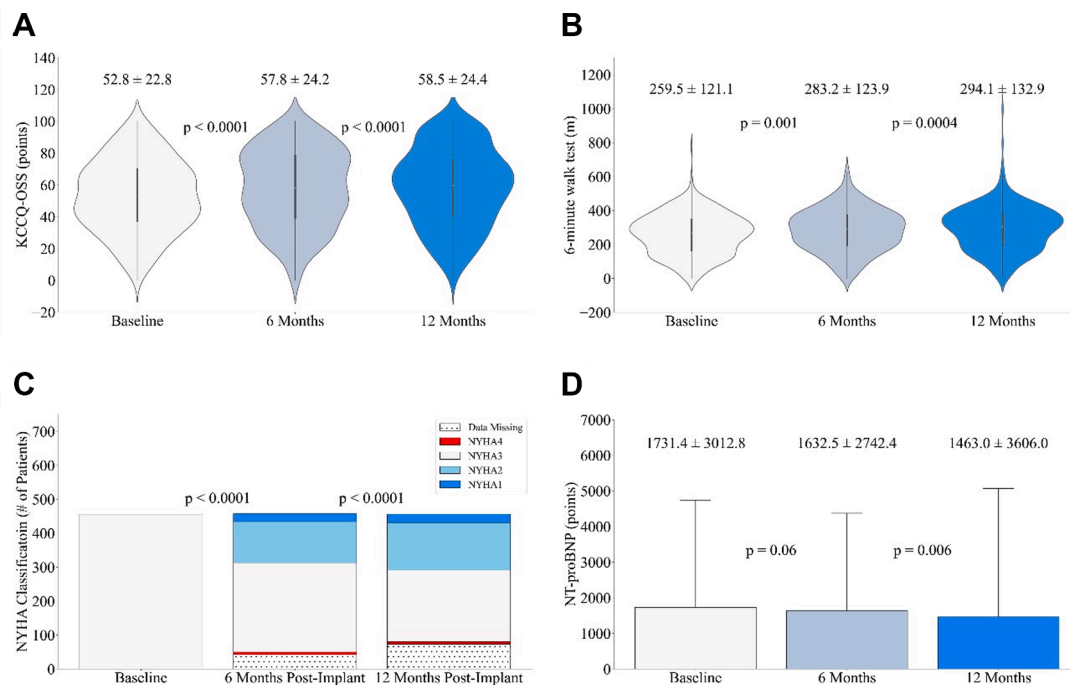


There were 162 (78%) HFrEF patients with paired echocardiogram at baseline and 12 months. At baseline, the average LVEF in patients with HFrEF was $26.4\% \pm 8.1\%$ compared with $32.0\% \pm 12.7\%$ at 12 months (paired sample Student's *t*-test $P < 0.0001$). Of these, 30 patients (19%) had HF with improved ejection fraction, defined as a baseline LVEF $\leq 40\%$, a ≥ 10 -point increase from baseline, and a second measurement of LVEF $> 40\%$. There were 172 (86%) patients with HFpEF with paired

echocardiogram at baseline and 12 months. At baseline, the average LVEF in HFpEF patients was $59.1\% \pm 6.6\%$ compared with $58.0\% \pm 8.4\%$ at 12 months (paired sample Student's *t*-test $P = 0.07$). Of these, 7 (4%) worsened to HFrEF (Supplemental Figure 4).

For patients above target seated mPAP > 20 mm Hg at baseline ($N = 206$), the seated trend mPAP decreased significantly over 12 months (29.2 ± 7.7 to 27.8 ± 9.9 mm Hg; $P = 0.04$). This decrease was more pronounced in patients with mPAP > 30 mm Hg at baseline

FIGURE 2 Functional and Biomarkers Changes



(A) Violin plot of paired analysis of KCCQ-OSS from baseline through 6 and 12 months. (B) Violin plot of paired analysis of 6MWT from baseline through 6 and 12 months. (C) Change in NYHA functional classification from baseline through 6 and 12 months. (D) Paired analysis of NT-proBNP from baseline through 6 and 12 months. All P values are compared with to baseline. KCCQ-OSS = Kansas City Cardiomyopathy Questionnaire-Overall Summary Score; NT-proBNP = N-terminal pro-B-type brain natriuretic peptide.

(37.2 ± 6.7 to 29.8 ± 11.0 ; $P < 0.01$). For patients with seated mPAP ≤ 20 mm Hg at baseline ($N = 247$), the seated trend mPAP increased significantly, while, on average, staying within target range (12.8 ± 4.9 mm Hg to 18.4 ± 7.4 mm Hg, paired sample Student's t -test $P < 0.0001$). Overall, seated trend mPAP increased significantly over 12 months (20.2 ± 10.3 mm Hg to 22.4 ± 9.7 mm Hg, paired sample Student's t -test $P < 0.0001$). Similarly, in both HFpEF and HFrEF patients, overall seated trend mPAP increased significantly over 12 months (HFpEF: 20.6 ± 10.2 mm Hg vs 23.5 ± 10.1 mm Hg; $P < 0.0001$; HFrEF: 19.9 ± 10.6 mm Hg vs 21.5 ± 9.7 mm Hg; $P = 0.006$). However, seated mPAP decreased significantly over 12 months for patients with baseline seated mPAP > 30 mm Hg at baseline across the ejection fraction spectrum (HFpEF: 38.5 ± 8.0 mm Hg vs 33.3 ± 11.2 mm Hg; $P = 0.005$; HFrEF: 37.2 ± 6.0 mm Hg vs 26.8 ± 10.6 mm Hg; $P < 0.0001$) (Supplemental Figure 5).

There were frequent outpatient HF medication changes made in the first 12 months (1.1 ± 1.4 changes per month per patient) with increases occurring slightly but significantly more frequently than

decreases (0.6 ± 0.8 vs 0.5 ± 0.7 , Mann-Whitney U $P = 0.04$). In total, there were 5,793 HF medication changes with 3,981 (69%) oral diuretic changes and 1,553 (27%) GDMT changes. There were 3,120 medication increases and 2,673 decreases ($P < 0.001$). GDMT includes renin-angiotensin system inhibitors, mineralocorticoid receptor antagonists, beta blockers, and sodium-glucose cotransporter 2 (SGLT2) inhibitors. The number of changes declined over time with 3,119 changes in the first 6 months compared with 2,674 changes in months 7 to 12, Mann-Whitney U $P < 0.001$. Patients above target mPAP at baseline had more frequent changes than those below/within target range (1.4 ± 1.7 vs 0.9 ± 1.1 changes per month, Mann-Whitney U $P = 0.001$) (Supplemental Figure 6). Diuretics were the most frequently adjusted medication. Although the same proportion of patients were on diuretic agents at baseline and 12 months (92%), there was a significant increase in the dose at 12 months (136 ± 153 mg vs 176 ± 215 mg furosemide equivalents, Mann-Whitney U $P < 0.001$). Overall, 165 (40.9%) had an increase in diuretic dose over 12 months, 100 (24.8%) had a

decrease, and 138 (34.2%) remained stable. Although the same proportion of patients were on GDMT at baseline and 12 months, there was an increase in the dose at 12 months in angiotensin receptor-neprilysin inhibitor (171 ± 127 mg vs 197 ± 142 mg, Mann-Whitney U $P = 0.03$) and aldosterone antagonist (31 ± 21 mg vs 36 ± 29 mg, Mann-Whitney U $P = 0.01$). In HFpEF and HFrEF patients, the proportion of patients on all HF medications was the same at baseline and 12 months. There were significantly more changes in patients with HFpEF than in patients with HFrEF (14.5 vs 10.7 changes per patient, Mann-Whitney U $P = 0.001$). For patients with HFpEF, there was an increase in the dose at 12 months in angiotensin receptor-neprilysin inhibitor (144 ± 107 mg vs 200 ± 180 mg, Mann-Whitney U $P = 0.03$), loop diuretic (168 ± 180 mg vs 220 ± 270 mg, Mann-Whitney U $P < 0.001$), and aldosterone antagonist (36 ± 27 mg vs 41 ± 37 mg, Mann-Whitney U $P = 0.03$). For patients with HFrEF, there was an increase in the dose at 12 months in loop diuretic (106 ± 106 mg vs 136 ± 144 mg, Mann-Whitney U $P < 0.001$). For patients with HFrEF, the baseline GDMT score was 6.9 vs 6.7 at 12 months. Patients above target range seated mPAP at baseline had significantly more changes than patients below/within target range (15.7 vs 10.3 changes per patient, Mann-Whitney U $P = 0.001$) (Table 2, Supplemental Figure 7).

Average systolic blood pressure decreased significantly for in-office visits (121.8 ± 19.1 mm Hg vs 116.4 ± 17.8 mm Hg, paired sample Student's t -test $P < 0.0001$, paired $n = 388$) but not for at home (118.4 ± 20.0 mm Hg vs 116.3 ± 18.1 mm Hg, paired sample Student's t -test $P = 0.06$, paired $n = 370$) through 12 months. Average weight measured at home decreased significantly (104.7 ± 25.3 kg vs 103.7 ± 24.9 kg, paired sample Student's t -test $P = 0.001$, paired $n = 382$). Patients with HFpEF saw significant decreases in average systolic blood pressure for both in-office visits (128.1 ± 19.0 mm Hg vs 122.4 ± 17.8 mm Hg, paired sample Student's t -test $P = 0.002$) and at home (125.2 ± 20.0 mm Hg vs 121.2 ± 17.4 mm Hg, paired sample Student's t -test $P < 0.02$) and weight (106.9 kg ± 26.0 kg vs 105.2 ± 25.0 kg, paired sample Student's t -test $P = 0.002$). Patients with HFrEF patients saw a significant decrease in average systolic blood pressure for in-office visits (115.1 ± 17.1 mm Hg vs 110.6 ± 16.2 mm Hg, paired sample Student's t -test $P < 0.002$). All vital sign changes through 12 months can be found in Supplemental Tables 2 and 3.

Patient and clinic compliance at 12 months was high with average patient compliance being 85% of weeks compliant (≥ 5 readings) with an average of 5.9 submissions per week and clinics reaching the

acknowledgment target (acknowledge within 4 days) for 92% of readings with an average of 2.4 days to acknowledge each submission (see Supplemental Figure 8 for weekly breakdown of compliance).

Details on all other secondary endpoint results can be found in the Supplemental Tables 4 to 8.

There were 4 (0.8%) patients with device/system-related complications and there was 1 (0.2%) patient with pressure sensor failure with none occurring after the 6-month primary endpoint. Serious adverse events were low with 12 (2.6%) patients experiencing acute kidney injury (Supplemental Tables 9 to 11).

DISCUSSION

In PROACTIVE-HF, remote titration of diuretic agents and GDMT in patients with NYHA functional class III HF on stable drug therapy, with the intent of optimizing seated mPAP and vital signs to improve outcomes over 12 months, was effective. The powered secondary endpoint of 12-month incidence of HF hospitalization and all-cause mortality was 0.36 in the primary cohort, which was nearly half of the performance goal set from previous hemodynamic-guided trials. The only group that failed to surpass the performance goal was the former control arm patients, who during the randomized period had significantly more events than the former treatment arm.⁷ Although the single-arm design was not powered to compare the arms of the former randomized trial cohort, it is reasonable to speculate that these differences were indeed driven by clinician access to seated mPAP. In the 12 months following unblinding the former control arm to PAP, event rates were comparable to the results of the primary cohort (0.27 vs 0.36 events/patient/12-month).⁹

Compared with previous PAP-guided HF management studies in patients with HF with NYHA functional class III symptoms (0.52-0.73 events per patient per 12-month) in both randomized and single cohort trials,¹²⁻¹⁶ PROACTIVE-HF showed markedly low rates of HF hospitalization and death at 12 months and these event rates are consistent with previously reported Cordella event rates in both the United States and Europe.^{9,17}

Patients with seated mPAP above target range at baseline saw a significant reduction in mPAP (-1.4 mm Hg) and patients with within/below target range saw a significant increase in mPAP ($+5.6$ mm Hg), although, on average, these patients remained within target mPAP range at 12 months. These trends (significant decrease for high PAP and significant increase for low PAP) were observed across the ejection fraction spectrum. Remote HF

TABLE 2 Baseline and 12-Month HF Medication by Class (N = 456)

	Baseline		12-Month		P Value	
	Count	Dose	Count	Dose	Count	Dose
ARNI	200 (43.9)	171.3 ± 127.2	180 (44.0)	196.5 (142.3)	0.96	0.03
ARB	83 (18.2)	32.8 ± 24.7	77 (18.8)	30.0 (24.0)	0.81	0.37
ACEI	26 (5.7)	14.2 ± 12.3	21 (5.1)	12.3 (12.2)	0.71	0.07
Beta-blocker	387 (84.9)	51.5 ± 186.6	336 (82.2)	39.0 (118.8)	0.28	0.23
Loop diuretic	417 (91.5)	135.6 ± 152.6	377 (92.2)	176.3 (215.1)	0.70	<0.001
Thiazide diuretic	72 (15.8)	3.0 ± 2.0	82 (20.1)	3.2 (2.8)	0.10	0.04
Aldosterone antagonist	307 (67.3)	30.5 ± 21.3	277 (67.7)	36.2 (29.2)	0.90	0.01
SGLT2 inhibitor	261 (57.2)	12.0 ± 5.7	237 (58.0)	12.4 (6.3)	0.83	0.23
Hydralazine	35 (7.7)	150.9 ± 112.8	36 (8.8)	156.6 (120.2)	0.55	0.43
Nitrates	88 (19.3)	34.3 ± 46.9	80 (19.6)	29.7 (45.7)	0.92	0.64
HFpEF subjects (n = 201)						
ARNI	25 (12.4)	143.7 ± 107.2	30 (16.6)	200.2 ± 180.4	0.25	0.03
ARB	48 (23.9)	39.3 ± 25.9	44 (24.3)	36.2 ± 26.9	0.92	0.53
ACEI	15 (7.5)	17.5 ± 13.4	13 (7.2)	13.9 ± 13.5	0.92	0.06
Beta-blocker	151 (75.1)	64.8 ± 199.6	133 (73.5)	55.2 ± 164.2	0.71	0.47
Loop diuretic	185 (92.0)	168.3 ± 179.7	168 (92.8)	220.4 ± 270.1	0.77	<0.001
Thiazide diuretic	41 (20.4)	2.9 ± 2.0	46 (25.4)	2.9 ± 2.0	0.24	0.10
Aldosterone antagonist	116 (57.7)	35.5 ± 26.9	114 (63.0)	41.4 ± 36.9	0.29	0.03
SGLT2 inhibitor	93 (46.3)	12.1 ± 6.0	87 (48.1)	13.0 ± 7.6	0.73	0.27
Hydralazine	23 (11.4)	153.5 ± 112.1	20 (11.1)	178.5 ± 127.6	0.90	0.73
Nitrates	37 (18.4)	37.3 ± 53.2	37 (20.4)	28.5 ± 39.4	0.62	0.83
HFrEF subjects (n = 208)						
ARNI	149 (71.6)	177.1 ± 132.8	127 (68.7)	196.6 ± 137.4	0.52	0.42
ARB	24 (11.5)	20.2 ± 17.6	27 (14.6)	21.5 ± 17.3	0.37	0.38
ACEI	10 (4.8)	7.8 ± 6.8	7 (3.8)	6.8 ± 6.1	0.62	0.72
Beta-blocker	196 (94.2)	34.5 ± 130.8	166 (89.7)	28.5 ± 80.8	0.10	0.52
Loop diuretic	191 (91.8)	105.6 ± 106.4	172 (93.0)	136.0 ± 143.9	0.67	<0.001
Thiazide diuretic	24 (11.5)	3.2 ± 2.3	27 (14.6)	3.0 ± 2.0	0.37	0.51
Aldosterone antagonist	153 (73.6)	27.6 ± 17.2	130 (70.3)	32.6 ± 22.6	0.47	0.13
SGLT2 inhibitor	142 (68.3)	11.9 ± 5.5	127 (68.7)	12.1 ± 5.6	0.94	0.51
Hydralazine	11 (5.3)	157.3 ± 117.6	15 (8.1)	117.8 ± 101.2	0.26	0.55
Nitrates	43 (20.7)	32.2 ± 42.9	38 (20.5)	27.8 ± 41.0	0.97	0.39

Values are n (%) or mean ± SD, unless otherwise indicated. **Bold** indicates $P < 0.05$.
Abbreviations as in [Table 1](#).

management of seated PA pressures is a novel feature of the Cordella system. Prior PAP management studies and technology used supine PAP for remote management of HF and clinician intuition of patient volume status vis-à-vis PAP evolved in the era of supine right heart catheterization. However, when moving from a supine to a seated position, blood volume shifts from the thorax to below the diaphragm under gravitational force and, in patients with HF, the ability to compensate for this shift in volume and recruit preload reserve may vary based on differing splanchnic and peripheral capacitance, which is masked in the supine position as venous pooling does not occur.^{7,18,19} The overall increase in PAP observed in the population may have been related to lack of familiarity with orthostatic PAP physiology, which may have influenced clinician

decision making, particularly in the patients below/within seated mPAP goals. Given the seated mPAP goal was 5 to 20 mm Hg, subjects with rising PAPs that remained within target range likely did not receive intervention, leading to an overall increase for subjects with baseline PAP ≤20 mm Hg. Furthermore, there may have been a minor contribution toward upward bias of measurement, as Cordella reported a +1.4 mm Hg mPAP difference from reference right heart catheterization at 90 days whereas CardioMEMS (CardioMEMS HF System) previously reported a −1.6 mm Hg downward bias at 60 days.^{20,21}

Although neurohormonal modulators have been shown to be effective in reducing HF hospitalization, the response varies across the ejection fraction spectrum, with the therapeutic benefit primarily

seen in patients with HFrEF.²²⁻²⁴ A recent analysis of GUIDE-HF (Hemodynamic-GUIDEd management of Heart Failure) sought to examine whether hemodynamically guided management of patients with HF based on direct measurement of filling pressures was effective in reducing morbidity and mortality across the ejection fraction spectrum.^{5,12} The authors found that this strategy was effective in reducing HF hospitalizations across the ejection fraction spectrum in patients with NYHA functional class II-III enrolled with a previous HF hospitalization or increased natriuretic peptides. In PROACTIVE-HF, patients were on the highest baseline use of GDMT including angiotensin receptor-neprilysin inhibitor (73% in HFrEF) and SGLT2 inhibitor (56.8%) among all hemodynamic monitoring studies to date (performance goal was adjusted down for an expected 50% SGLT2 inhibitor utilization). Over 12 months, the proportion of patients on GDMT did not change. However, there were significant increases in the average dose of angiotensin receptor-neprilysin inhibitor, loop and thiazide diuretic, and aldosterone antagonist, suggesting that the Cordella system enabled further titration of GDMT in an HF population on stable GDMT at baseline. Patients with HFpEF saw an increase in the average dose of angiotensin receptor-neprilysin inhibitor, loop diuretics, and aldosterone antagonists whereas patients with HFrEF saw an increase in the average dose of loop diuretic agents and there was no change in GDMT score through 12 months. Although the direct relationship between different medication titrations and resulting PAP change warrants further study, the seated mPAP trends suggest that clinicians were following the treatment guidelines to target a seated mPAP 5-20, primarily through uptitrating diuretic agents in patients with high PAP at baseline, resulting in low rates of HF hospitalizations across the ejection fraction spectrum.²⁵

The single-arm design of PROACTIVE-HF warrants careful interpretation of the results. Nevertheless, there were significant improvements across several key measures: KCCQ (+5.7 points), 6MWT (+34.5 m), NT-proBNP (−268.4 pg/mL), NYHA functional class (43% improved), systolic blood pressure (−5.4 mm Hg), weight (−1 kg), and improved ejection fraction in patients with HFrEF over 12 months. These results provide additional evidentiary context for the effectiveness of the Cordella system in managing patients with NYHA functional class III HF with a previous HF hospitalization or increased natriuretic peptide levels. The combination of

monitoring true preload (seated PAP), afterload, and weight, with a system that enables high patient compliance and responsive clinics, allowed for timely intervention to reduce risk by lowering high PAP to prevent uncontrolled congestion, agnostic to LVEF.²⁶

STUDY LIMITATIONS. The major limitation of PROACTIVE-HF is its open-label, single-arm design and the absence of a concurrent control arm. Given the risk of bias due to absence of randomization, conclusions of the study should be interpreted cautiously. However, the endpoint assessment was blinded and performed by an independent endpoint adjudication committee and the differences in the event rates of the former control and treatment arms were indicative of a clinically and statistically significant difference. Although KCCQ-Overall Summary Score improved significantly, there are known limitations of interpretation of KCCQ in unblinded studies. There was no core laboratory evaluation for ejection fraction classification and determination of ejection fraction may differ across the different clinical sites.

CONCLUSIONS

In summary, for HF patients with NYHA functional class III symptoms at risk of congestive events, the Cordella PA Sensor and HF System was safe, improved quality of life and functional capacity, and enabled a significant reduction in PAP in patients with elevated baseline seated mean PAP. The integration of daily vital signs with seated mPAP, high patient and clinician engagement, and high rates of GDMT at baseline enabled markedly low rates of HF hospitalization and all-cause mortality across the ejection fraction spectrum through 12 months.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

This study was funded by Endotronix Inc, Chicago, Illinois, USA. Drs Miranda, Hickey, Shah, and Macaluso were on the Eligibility Committee of PROACTIVE-HF. Prof Sharif is funded by the Science Foundation Ireland SFI 17/RI/5353. Drs Sauerland, Forouzan, Owens, and Hiivala are employees of Endotronix. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

ADDRESS FOR CORRESPONDENCE: Dr Liviu Klein, Advanced Heart Failure Comprehensive Care Center and Division of Cardiology, University of California-San Francisco, M-1178B, Box 0124, 505 Parnassus Avenue, San Francisco, California 94143, USA. E-mail: Liviu.Klein@ucsf.edu.

PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: Over 12 months of remote monitoring of daily seated mPAP in conjunction with heart rate, blood pressure, and weight is safe and effective in achieving low rates of HF hospitalizations through 12 months in patients with chronic heart failure.

TRANSLATIONAL OUTLOOK: Understanding the orthostatic-supine physiology of PAP changes across the

ejection fraction spectrum may provide better differential diagnosis in patients with symptomatic HF in the outpatient setting. Remote monitoring of several physiologic parameters (PAP, blood pressure, weight, and heart rate) simultaneously may provide better guidance, across the ejection fraction, for treating patients with chronic HF.

REFERENCES

- Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2022;79(17):e263–e421.
- Zile MR, Bennett TD, Hajj SE, et al. Intracardiac pressures measured using an implantable hemodynamic monitor: relationship to mortality in patients with chronic heart failure. *Circ Heart Fail*. 2017;10:e003594.
- Stevenson LW, Ross HJ, Rathman LD, Boehmer JP. Remote monitoring for heart failure management at home. *J Am Coll Cardiol*. 2023;81:2272–2291.
- Redfield MR, Borlaug BA. Heart failure with preserved ejection fraction: a review. *JAMA*. 2023;329:827–838.
- Zile MR, Mehra MR, Ducharme A, et al. Hemodynamically-guided management of heart failure across the ejection fraction spectrum: the GUIDE-HF trial. *JACC Heart Fail*. 2022;10:931–944.
- Lindenfeld J, Costanzo MR, Zile MR, et al. Implantable hemodynamic monitors improve survival in patients with heart failure and reduced ejection fraction. *J Am Coll Cardiol*. 2024;83:682–694.
- Guichard JL, Bonno EL, Nassif ME, et al. Seated pulmonary artery pressure monitoring in patients with heart failure: results of the PROACTIVE-HF trial. *JACC Heart Fail*. 2024;12:1879–1893.
- Guichard JL, Cowger JA, Chaparro SV, et al. Rationale and design of the PROACTIVE-HF trial for managing patients with NYHA class III heart failure by using the combined Cordella pulmonary artery sensor and the Cordella heart failure system. *J Card Fail*. 2023;29:171–180.
- Cowger JA, Guichard JL, Miranda D, et al. Engaging patients and clinicians with remote pulmonary artery pressures improves care: a substudy of the PROACTIVE-HF trial. *J Card Fail*. 2024;30(1):226.
- Guichard JL, Sharif F, Forouzan O, Martina J, Klein L. A procedural guide for implanting the Cordella pulmonary artery pressure sensor. *J Invasive Cardiol*. 2023;35:E75–E83.
- Kovacs G, Berghold A, Scheidl S, Olschewski H. Pulmonary arterial pressure during rest and exercise in healthy subjects: a systematic review. *Eur Respir J*. 2009;34:888–894.
- Lindenfeld J, Zile MR, Desai A, et al. Haemodynamic-guided management of heart failure (GUIDE-HF): a randomised controlled trial. *Lancet*. 2021;398:991–1001.
- Abraham WT, Adamson PB, Bourge RC, et al. Wireless pulmonary artery haemodynamic monitoring in chronic heart failure: a randomized controlled trial. *Lancet*. 2011;377:658–666.
- Brugts JJ, Radhoe SP, Clephas PRD, et al. Remote haemodynamic monitoring of pulmonary artery pressures in patients with chronic heart failure (MONITOR-HF): a randomised clinical trial. *Lancet*. 2023;401:2113–2123.
- Shavelle DM, Desai AK, Abraham WT, et al. Lower rates of heart failure and all-cause hospitalizations during pulmonary artery pressure-guided therapy for ambulatory heart failure: one-year outcomes from the CardioMEMS post-approval study. *Circ Heart Fail*. 2020;13:e006863.
- Angermann CE, Assmus B, Anker SD, et al. Pulmonary artery pressure-guided therapy in ambulatory patients with symptomatic heart failure: the CardioMEMS European Monitoring Study for Heart Failure (MEMS-HF). *Eur J Heart Fail*. 2020;22:1891–1901.
- Sharif F, Rosenkrantz S, Bartunek J, et al. Twelve-month follow-up results from the SIRONA 2 clinical trial. *ESC Heart Fail*. 2024;11:1133–1143.
- Fudim M, Sobotka PA, Dunlap ME. Extracardiac abnormalities of preload reserve: mechanisms underlying exercise limitation in heart failure with preserved ejection fraction, autonomic dysfunction, and liver disease. *Circ Heart Fail*. 2021;14:e007308.
- Bronzwaer ASGT, Bogert LWJ, Westerhof BE, Piek JJ, Daemen MJAP, van Lieshout JJ. Abnormal haemodynamic postural response in patients with chronic heart failure. *ESC Heart Fail*. 2017;4:146–153.
- Sharif F, Rosenkrantz S, Bartunek J, et al. Safety and efficacy of a wireless pulmonary artery pressure sensor: primary endpoint results of the SIRONA 2 clinical trial. *ESC Heart Fail*. 2022;9:2862–2872.
- Abraham WT, Adamson PB, Hasan A, et al. Safety and accuracy of a wireless pulmonary artery pressure monitoring system in patients with heart failure. *Am Heart J*. 2011;161:558–566.
- Dewan P, Jackson A, Lam CSP, et al. Interactions between left ventricular ejection fraction, sex and effect of neurohormonal modulators in heart failure. *Eur J Heart Fail*. 2020;22:898–901.
- Butler J, Packer M, Filippatos G, et al. Effect of empagliflozin in patients with heart failure across the spectrum of left ventricular ejection fraction. *Eur Heart J*. 2022;43:416–426.
- Solomon SD, Vaduganathan M, Claggett BL, et al. Sacubitril/Valsartan across the spectrum of ejection fraction in heart failure. *Circulation*. 2020;141:352–361.
- Zile MR, Bennett TD, Hajj SE, et al. Intracardiac pressures measured using an implantable hemodynamic monitor: relationship to mortality in patients with chronic heart failure. *Circ Heart Fail*. 2017;10:e003594.
- Adamson PB. Pathophysiology of the transition from chronic compensated and acute decompensated heart failure: new insights from continuous monitoring devices. *Curr Heart Fail Rep*. 2009;6:287–292.

APPENDIX For study organization lists and an expanded Methods section as well as supplemental figures and tables, please see the online version of this paper.



Go to <https://www.acc.org/jacc-journals-cme> to take the CME/MOC quiz for this paper.